

Generation Interconnection System Impact Study Report

For Queue Project Q17-0869

New Hollandale 138 kV

September 23, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0869. The Interconnection Customer proposes a 113.1 MW solar photovoltaic facility interconnecting at the New Hollandale 138 kV substation in the Ohio area. The request was received on October 03, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 389 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 18 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,394,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: BINGEN to CENTRAL 138 KV Line
Contingency: WAGNER LAKE to OSAWATOMIE 230 KV line
Loading: 149.3% (Rating: 217 MVA)
DFAX: 40.1%
Pre-project loading: 128.4%
2. Facility: CENTRAL to REDFORD 138 KV
Contingency: WAGNER LAKE to OSAWATOMIE 230 KV Line
Loading: 148.8% (Rating: 228 MVA)
DFAX: 40.1%
Pre-project loading: 128.9%
3. Facility: REDFORD to NEW HOLLANDALE 138 KV
Contingency: WANER LAKE to OSAWATOMIE 230 KV line
Loading: 148.2% (Rating: 241 MVA)
DFAX: 40.1%
Pre-project loading: 129.4%
4. Facility: WASHINGTON COUNTY COGENERATION FACILITY to MOSSYROCK 230 KV Line
Contingency: BARRETT to WOLFE 500 KV
Loading: 128.7% (Rating: 928 MVA)
DFAX: 12.4%
Pre-project loading: 127.2%
5. Facility: NEW HOLLANDALE to WAGNER LAKE 230 KV line
Contingency: OAKLAND AVE to BAY POINT 230 KV
Loading: 124.8% (Rating: 469 MVA)
DFAX: 25.6%
Pre-project loading: 118.7%
6. Facility: NEW HOLLANDALE SUBSTATION 100-1 kV Transformer Bank
Contingency: REDFORD to NEW HOLLANDALE 138 KV
Loading: 122.0% (Rating: 126 MVA)

DFAX: 11.1%

Pre-project loading: 112.0%

7. Facility: OSAWATOMIE 100-1 kV Transformer

Contingency: BLUESTEM to BODO 230 KV

Loading: 118.0% (Rating: 96 MVA)

DFAX: 5.6%

Pre-project loading: 111.5%

8. Facility: RUTLAND 100/1kV Transformer Bank

Contingency: REDFORD to NEW HOLLANDALE 138 KV L

Loading: 112.9% (Rating: 100 MVA)

DFAX: 7.3%

Pre-project loading: 104.7%

9. Facility: WISSOTA to BERRY ST 138 KV

Contingency: DAVISVILLE to NORTH APPLETON 138 KV Line

Loading: 112.4% (Rating: 648 MVA)

DFAX: 8.4%

Pre-project loading: 111.0%

10. Facility: CHILHOWEE to OSAWATOMIE 100 KV

Contingency: OSAWATOMIE 100 kV/1 kV Auto

Loading: 108.7% (Rating: 112 MVA)

DFAX: 8.0%

Pre-project loading: 100.7%

11. Facility: WISSTA to SAN ANDREAS 138 KV

Contingency: WISSOTA to NSP 138 KV L

Loading: 104.2% (Rating: 383 MVA)

DFAX: 5.3%

Pre-project loading: 102.7%

12. Facility: FLATLICK to COMBEE RD SUBSTATION 230 KV Line

Contingency: NORWICH to OMNI-HACKBERRY 500 KV

Loading: 103.7% (Rating: 767 MVA)

DFAX: 5.2%

Pre-project loading: 102.9%

13. Facility: MINCO to LARSON ST SUBSTATION 230 KV

Contingency: BARRETT to NORWICH 500 KV line

Loading: 102.9% (Rating: 1,179 MVA)

DFAX: 16.1%

Pre-project loading: 101.3%

14. Facility: CLAUS to UNION 138 KV

Contingency: WISSOTA to NSP 138 KV

Loading: 102.6% (Rating: 367 MVA)

DFAX: 5.3%

Pre-project loading: 101.0%

15. Facility: CLAUS to SAN ANDREAS 138 KV

Contingency: WISSOTA to NSP 138 KV

Loading: 101.8% (Rating: 388 MVA)

DFAX: 5.3%

Pre-project loading: 100.3%

16. Facility: WASHINGTON COUNTY COGENERATION FACILITY to VILLA PARK 230 KV

Contingency: WOLFE to BARRETT 500 KV

Loading: 101.5% (Rating: 1,057 MVA)

DFAX: 13.0%

Pre-project loading: 100.1%

17. Facility: WOLFE to VILLA PARK 230 KV

Contingency: WOLFE to BARRETT 500 KV

Loading: 101.0% (Rating: 1,094 MVA)

DFAX: 13.0%

Pre-project loading: 99.7%

18. Facility: GREENLY to DRESDEN 500 KV line

Contingency: INTERNATIONAL PAPER to DRESDEN 500 KV Line

Loading: 100.8% (Rating: 3,238 MVA)

DFAX: 23.2%

Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BINGEN to CENTRAL 138 KV Line	Rebuild line with higher-capacity conductor	\$1,475,000	100% (\$1,475,000)
CENTRAL to REDFORD 138 KV	Rebuild line with higher-capacity conductor	\$1,424,000	100% (\$1,424,000)
REDFORD to NEW HOLLANDALE 138 KV	Rebuild line with higher-capacity conductor	\$1,063,000	100% (\$1,063,000)
WASHINGTON COUNTY COGENERATION FACILITY to MOSSYROCK 230 KV	Rebuild line with higher-capacity conductor	\$52,041,000	3% (\$1,368,000)
WAGNER LAKE to NEW HOLLANDALE 230 KV	Rebuild line with higher-capacity conductor	\$13,345,000	25% (\$3,357,000)
NEW HOLLANDALE SUBSTATION 100-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$3,274,000	46% (\$1,497,000)
OSAWATOMIE 100-1 kV Transformer	Replace transformer with higher-rated unit	\$4,731,000	100% (\$4,731,000)
RUTLAND 100-1 kV Auto	Replace transformer with higher-rated unit	\$2,768,000	100% (\$2,768,000)
WISSOTA to BERRY ST 138 KV	Rebuild line with higher-capacity conductor	\$1,296,000	4% (\$55,000)
CHILHOWEE to OSAWATOMIE 100 KV	Reconductor line	\$331,000	73% (\$241,000)
WISSTA to SAN ANDREAS 138 KV	Reconductor line	\$2,519,000	4% (\$112,000)
COMBEE RD to FLATLICK 230 KV L	Reconductor line	\$10,872,000	2% (\$256,000)
MINCO to LARSONSTREET 230 KV	Reconductor line	\$4,258,000	2% (\$75,000)
CLAUS to UNION 138 KV	Reconductor line	\$628,000	4% (\$28,000)
SAN ANDREAS to CLAUS 138 KV Line	Reconductor line	\$2,405,000	4% (\$107,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WASHINGTON COUNTY COGENERATION FACILITY to VILLA PARK 230 KV	Reconductor line	\$16,204,000	2% (\$377,000)
WOLFE to VILLA PARK 230 KV Line	Reconductor line	\$6,015,000	2% (\$140,000)
GREENLY to DRESDEN 500 KV	Reconductor line	\$28,466,000	1% (\$320,000)
	Total allocated to project		\$19,394,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.